1. In an approximation, the difference between the exact solution (x_{true}) and the approximate solution ($x_{approximate}$) is called the *true error* and its formula is: $E_t = x_{true} - x_{approximate}$. However, you have learnt a second type of error: *relative error* ($E_r = \left|\frac{x_{true} - x_{approximate}}{x_{true}}\right|$). Why might we need to use relative error? Please explain with a simple example that we use relative error in numerical calculations to show that the approximation is more meaningful/better. (5p)

2. Approximate the function $f(x) = \boldsymbol{sinx}$ by using Taylor series expansion about $x = 0$, using two, four, and six terms. In each case, calculate the approximate value of the function at $x = \frac{\pi}{2}$. (20p)

3. Consider the equation $x^3 - x - 1 = 0$.
   a. Prove that this equation has at least one real root in the interval [1,2](5p).
   b. Calculate the approximate root of the given equation within the interval [1,2] using the Bisection method after two iterations. (10p)

4. Consider the equation $e^x + x = 7$. Use Newton's method to approximate the solution to 4 correct digits. (15p)

5. Consider this function:

$$f(x) = x^3 - x - e^x - 2$$

An approximate solution will be calculated using the fixed-point iteration method with the initial condition $x_0 = 2$.

a) For this function; find the iteration formula using the initial condition. (Note: This function has 4 different iteration formula candidates.)(20p)

b) Using this iteration formula, calculate the approximate root after 2 iterations. (10p)

6. Apply the Jacobi method to solve the following system of linear equations:

$$\begin{aligned} 4x_1 - x_2 + x_3 &= 7 \\ 4x_1 - 8x_2 + x_3 &= -21 \\ -2x_1 + x_2 + 5x_3 &= 15 \end{aligned}$$

Apply two iterations and write down the approximate values of x_1, x_2, x_3. (15p)

① Relative error → $\frac{|x - \tilde{x}|}{x}$

scenario	x_{true}	x_{Apr}
A	1	2
B	100	101

$Er_A = \frac{|1-2|}{1} \rightarrow 1$

$Er_B = \frac{|100-101|}{100} \rightarrow 0.01$

② $f(x) \approx f(x_0) + f'(x_0).(x-x_0) + \frac{f''(x_0).(x-x_0)^2}{2!} + \frac{f'''(x_0).(x-x_0)^3}{3!}$

$+ \frac{f^{IV}(x_0).(x-x_0)^4}{4!} + \frac{f^{V}(x_0).(x-x_0)^5}{5!}$

$f'(x) \to \sin x' \to \cos x$
$f''(x) \to \cos x' \to -\sin x$
$f'''(x) \to -\sin x \to -\cos x$
$f^{IV}(x) \to -\cos x \to \sin x$
$f^{5}(x) \to \sin x \to \cos x$

a) two terms

$f(x) \approx \cancel{\sin 0} + \overset{1}{\cancel{\cos 0}}.(x-0) \to x \to \frac{\pi}{2}$

b) four terms

$f(x) \approx \cancel{\sin 0} + \cos 0.(x-0) + \frac{-\sin 0.\cancel{(x)^2}}{\cancel{2!}} + \frac{-\cos 0.x^3}{3!}$

$x - \frac{x^3}{6} \to \frac{\pi}{2} - \frac{(\frac{\pi}{2})^3}{6} = 0.924832$

c) six terms

$f(x) \approx \cancel{\sin 0} + \cos 0.(x-0) + \frac{-\sin 0.x^2}{\cancel{2!}} - \frac{\cos 0.x^3}{3!} + \frac{\cancel{\sin 0}.x^4}{\cancel{4!}} + \frac{\cos 0.x^5}{5!}$

$x - \frac{x^3}{6} + \frac{x^5}{120} \to 1.00452$

(3) (A)

$f(x) = x^3 - x - 1$ aralık $[1,2]$

$[1,2]$

$1^3 - 1 - 1 = -1 \rightarrow f(1)$

$2^3 - 2 - 2 = 4 \rightarrow f(2)$

> $f(1) \cdot f(2) < 0$ so at least one root

(B)

1. iterasyon

$\frac{1+2}{2} = 1.5$

$f(1.5) = (1.5)^3 - 1.5 - 1 = 0.875$

$f(1) \cdot f(1.5) < 0$ so new interval is $[1, 1.5]$

2. iterasyon

$\frac{1+1.5}{2} = 1.125$

(4) $e^x + x - 7 = 0$ Newton's method

$x=0 \rightarrow e^0 + 0 - 7 = 1 - 7 = -6$

$x=1 \rightarrow e + 1 - 7 = -3.2817$

$x=2 \rightarrow e^2 + 2 - 7 = 1.389$

> $f(1) \cdot f(2) < 0$ so there is at least one root

$[1,2] \rightarrow x_0 = 1.5$ optimal

Newton

$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ $\quad i=0 \rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$f(1.5) = -1.01831$

$f'(x) = e^x + 1$

$f'(1.5) = e^{1.5} + 1 = 5.4816$

$x_1 = 1.5 + \frac{1.01831}{5.4816}$

$= 1.68576$

⑤ $f(x) = x^3 - x - e^x - 2$ $\quad x_0 = 2$

1st candidate

$$x^3 = x + e^x + 2$$

$$x = \sqrt[3]{x + e^x + 2}$$

$$g(x) = \sqrt[3]{x + e^x + 2}$$

2nd candidate

$$x = x^3 - e^x - 2$$

$$g(x) = x^3 - e^x - 2$$

3rd candidate

$$e^x = x^3 - x - 2$$

$$x = \ln(x^3 - x - 2)$$

$$g(x) = \ln(x^3 - x - 2)$$

4th candidate

$$x(x^2 - 1) - e^x - 2$$

$$x = \frac{e^x + 2}{x^2 - 1}$$

$$g(x) = \frac{e^x + 2}{x^2 - 1}$$

$$g(x) = \sqrt[3]{x + e^x + 2}$$

$$g'(x) = \frac{1}{3} \cdot (x + e^x + 2)^{-2/3} \cdot (1 + e^x)$$

$$g'(2) = \frac{1}{3} \cdot (2 + e^2 + 2)^{-2/3} \cdot (1 + e^2)$$

$$= 0.55$$

$g'(2) < 1$ so we can use this formula

1st iteration

$$x_{i+1} = \sqrt[3]{x_i + e^{x_i} + 2}$$

$g(x) = x_{i+1}$

$x_0 = 2$

$i = 0$ $\quad x_1 = \sqrt[3]{2 + e^2 + 2} = 2.249896$

$i = 1$ $\quad x_2 = \sqrt[3]{2.249896 + e^{2.249896} + 2} = \underline{\underline{2.3949}}$

after two iterations approx. root is 2.3949